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JAVA PROJECT REPORT ON

Secure User Login and Authentication System with Password Validation

Submitted by

PRIYANKA S

25BIT041

Under the Guidance of

DHILEEBAN A

Technical Trainer

Technical Department, KG College of Arts and Science

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DECLARATION

I hereby declare that the project entitled "**Secure User Login and Authentication System with Password Validation"** submitted in partial fulfillment of the requirements for the course **[JAVA PROGRAMMING]** is a **record of my original work** carried out under the guidance of **[DHILEEBAN A]**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

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CERTIFICATE

This is to certify that the project entitled **"Secure User Login and Authentication System with Password Validation"** is a **bona fide work** carried out by **[PRIYANKA S] [B.Sc IT]**, [**Information Technology]**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **[JAVA PROGRAMMING]**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: COIMBATORE

Date:09/04/2026

Submitted for the Viva-Voce Examination held on _____

Guide:
(Technical Trainer, KGCAS)

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Head of the Department
Technical Head,(KGCAS)

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ABSTRACT

The **Secure User Login and Authentication System** is a Java-based console application designed to provide secure user authentication with robust password validation. In many software systems, weak authentication mechanisms can lead to unauthorized access and data breaches. This project aims to build a reliable login system that validates user credentials and enforces strong password policies.

The system allows users to register accounts with a username and password, log in securely, and manage their credentials. The application is developed using **Core Java programming concepts**, including classes, objects, methods, conditional statements, loops, and exception handling. The project can be implemented either as a **console-based application or using Java Swing for a graphical user interface**.

The main objective of the system is to prevent unauthorized access, enforce strong password policies, and provide a secure experience. The system validates credentials against stored records and provides clear feedback on authentication success or failure.

This project demonstrates the practical implementation of **object-oriented programming concepts in Java** and provides a basic foundation for developing more advanced secure application systems in the future.

1.INTRODUCTION

In modern software applications, user authentication is a critical component that ensures only authorized individuals can access protected resources. Traditionally, many systems used simple, unvalidated password schemes that made them vulnerable to unauthorized access. These weak authentication methods are time-consuming to maintain, prone to security breaches, and difficult to enforce consistently.

With the growth of digital systems, secure login and authentication mechanisms have become essential for protecting user data and system integrity. A well-designed authentication system enforces password rules, validates credentials accurately, and maintains organized records of registered users.

The **Secure User Login and Authentication System** is a Java-based application designed to provide a secure, console-based login and registration system. The system allows users to register with a unique username and a password that meets defined security criteria, log in with their credentials, and receive immediate feedback on authentication status.

This project is developed using **Core Java programming concepts** such as classes, objects, methods, conditional statements, loops, and exception handling. The application is implemented as a **console-based program with scope to extend to a Java Swing graphical user interface (GUI)**.

The primary goal of this system is to prevent unauthorized access, enforce strong password policies, and ensure secure credential management. This project also demonstrates how Java programming can be applied to develop secure, real-world authentication system.

1.1 Objective

The main objective of the **Secure User Login and Authentication System** is to develop a simple and efficient authentication system using Core Java that enforces secure login and password validation for software applications.

The specific objectives of this project are:

- To design and develop a **Java-based secure login and authentication system**.
- To implement **robust password validation with complexity rules**.

- To prevent **unauthorized access through strong credential verification** a **user-friendly registration and login interface** to register and authenticate securely.
- To demonstrate the implementation of **Core Java programming concepts** such as classes, objects, methods, loops, and conditional statements.
- To generate a **clear and informative authentication feedback to users**.

1.2 Overview

The **Secure User Login and Authentication System** is a console-based application developed using Java programming language. The system provides a secure digital platform where users can register accounts and log in using validated credentials.

The system prompts users to register with a username and password. The password must meet complexity requirements including minimum length, uppercase letters, digits, and special characters. Upon login, the system verifies the credentials and grants or denies access accordingly.

The system helps secure applications by enforcing password policies and rejecting invalid credentials. The application is implemented as a **console-based program with potential extension to a graphical user interface using Java Swing**.

This project demonstrates the practical use of Java programming in developing real-world secure authentication systems applicable to any software environment.

1.3 Features

The **Secure User Login and Authentication System** includes the following main features:

• **User Registration** – Allows new users to create an account by providing a unique username and a valid password.

• **Secure Login** – Authenticates users by verifying entered credentials against stored registration records.

• **Password Validation** – Enforces password rules including minimum length (8 characters), uppercase letters, digits, and special characters.

• **Duplicate Username Check** – Prevents registration of duplicate usernames to maintain unique user identities.

• **Authentication Feedback** – Provides clear messages for successful login, registration errors, and invalid credentials.

• **User-Friendly Console Interface** – Provides a simple, easy-to-use menu-driven console interface for all operations.

• **Password Change Option** – Allows authenticated users to update their password securely following validation rules.

2.PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should work and what features are required to solve the problem efficiently.

The analysis phase focuses on identifying the current issues in the existing system and proposing a computerized solution that improves efficiency and accuracy.

2.1 Problem Definition

In many software applications, authentication systems are poorly designed, using weak or unvalidated passwords that expose user data to security threats. This traditional approach creates several challenges.

Some common problems include:

• Weak or easily guessable passwords
• No enforcement of password complexity rules
• Risk of unauthorized access due to weak credentials
• No duplicate username detection
• Poor user feedback during authentication failures

Insecure authentication systems increase the risk of data breaches and erode user trust due to inadequate security measures.

Therefore, there is a need for an automated authentication system that can validate user credentials, enforce password policies, and manage user accounts securely.

The **Secure User Login and Authentication System** is designed to solve these problems by providing a secure, computerized solution that enforces password validation and manages user authentication.

2.2 Project Overview

The **Secure User Login and Authentication System** is a Java-based application developed to provide a secure and validated user login and registration experience.

The system provides a menu-driven console interface where users can register, log in, and manage their credentials. The system automatically validates passwords against defined security rules and stores user credentials for future authentication.

The system includes the following functionalities:

• Allowing new users to register with a unique username
• Validating passwords for complexity requirements
• Authenticating registered users at login
• Detecting duplicate usernames during registration
• Providing clear feedback for all authentication outcomes

This project is implemented using **Core Java programming concepts** and can be developed as either a **console-based program or a Java Swing GUI application**.

2.3 Scopes and Limitations

Scope

The scope of the **Secure User Login and Authentication System** includes the following:

• Registering new user accounts with validated credentials
• Authenticating users with stored credentials
• Enforcing password complexity policies
• Preventing duplicate username registrations
• Demonstrating the practical use of Java programming in security

The system can be used in:

• Web and desktop application login systems
• Corporate software systems
• Educational institution portals
• Banking and financial software
• Any Java-based multi-user application

3.SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The **Secure User Login and Authentication System** is designed as a menu-driven console application where users can register, log in, validate passwords, and manage their credentials securely.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system.

3.1 Modules

The main modules of the **Secure User Login and Authentication System** are explained below:

1. Menu Display Module

This module is responsible for displaying the system menu options to the user. The menu helps the user choose between registration, login, and other available functions.

System Menu Options:

1. Register (New user)
2. Login (Existing user)
3. Change Password
4. View Profile
5. Exit

The menu is displayed when the program starts so that the user can easily view the available options.

2. Order Selection Module

This module handles user registration. It accepts a username and password, checks for duplicate usernames, and validates the password before storing the new user record.

Functions of this module include:

• Accepting user input for item selection
• Validating password complexity rules
• Storing registered user credentials

The system ensures each username is unique and passwords meet all complexity requirements before confirming registration.

3. Bill Calculation Module

This module performs credential verification during login.

The system calculates:

• Total price of each item
• Username lookup in registered records
• Password matching verification
• Access grant or denial decision

The calculation is performed automatically using Java programming logic.

Example Calculation:

Item Total = Price × Quantity
if (storedPassword.equals(inputPassword)) → grant access
else → deny with error message

4. Invoice Generation Module

This module enforces password complexity rules before registration or password change.

Password rules include:

• Item name
• Minimum 8 characters
• At least one uppercase letter
• At least one digit (0-9)
• At least one special character (!@#$%)

Example Bill Output:

------ HOTEL BILL ------

--- LOGIN SUCCESSFUL ---

Welcome, john_doe!

--- REGISTRATION ERROR ---
Password must be at least 8 characters,
contain 1 uppercase, 1 digit, 1 special char.

Goodbye! Stay Secure!

5. Exit Module

This module allows the user to exit the application safely. On exit, a farewell security reminder is displayed.

When the user selects the exit option, the system terminates the program.

4.METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the **Secure User Login and Authentication System**, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

• Problem Analysis
• System Design
• Program Development
• Testing and Validation
• Result and Documentation

4.1. Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the needs of secure software authentication.

The main problems identified in insecure authentication systems are:

• Weak passwords pose a significant security risk.
• Lack of password validation leads to data breaches.
• No mechanism to prevent duplicate user registrations.
• Poor error messages provide no guidance to users.

Based on these problems, the system requirements were defined to develop a secure authentication application that enforces password policies and validates user credentials.

4.2. System Design

In this stage, the overall structure of the system is designed before coding begins.

The system is divided into modules such as:

• User Registration Module
• Login Authentication Module
• Password Validation Module
• Credential Storage Module
• Exit Module

The program flow and structure are designed so that users can easily register and log in with clear prompts and secure validation.

4.3. Program Development

After designing the system, the program is implemented using the **Java programming language**.

The following steps were followed during development:

• Creation of User class and authentication methods
• Implementation of menu-driven console logic
• Writing code for password validation logic
• Handling user input and registration data
• Displaying authentication success or failure messages

Core Java concepts such as **classes, objects, loops, conditional statements, and methods** were used during development.

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

Example Test Cases:

Test Case	**Expected Result**
Register with valid credentials	Registration successful, user stored
Register with weak password	Error message: password too weak
Login with correct credentials	Access granted, welcome message shown
Login with wrong password	Access denied, error message shown

This phase ensures that the program correctly validates passwords and authenticates users without errors.

4.5. Result and Documentation

After successful testing, the results of the system are documented. The system correctly registers users, validates passwords, and authenticates credentials.

The results are demonstrated through program execution and screenshots showing registration, login success, and error handling scenarios.

Proper documentation is prepared to explain the design, implementation, and working of the system.

4.6. Algorithm

Algorithm for the **Secure User Login and Authentication System**:

1. Start the program.
2. Display the main menu (Register / Login / Exit).
3. Ask the user to choose an option.
4. If Register: accept username and password input.
5. Validate the password against complexity rules.
6. If valid, store the user record; else show error.
7. If Login: accept username and password input.
8. Search stored records for matching credentials.
9. If found, grant access; else deny with error message.
10. Repeat menu until user selects Exit.
11. End the program.

4.7. Flowchart

Students should draw a flowchart showing the authentication flow.

Example flow:

Start
↓
Display Menu
↓
Select Item
↓
Enter Quantity
↓

Calculate Bill
↓
Display Invoice
↓
End

Students can draw using:

- draw.io
- Lucidchart
- MS Visio

6.SOURCE CODE

```
package com.securelogin;

import java.util.Scanner;

public class MainApp {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        UserServiceClass service = new UserServiceClass();

      while (true) {
            System.out.println("\n1. Register\n2.
Login\n3. Exit");
            System.out.print("Choose option: ");
            int choice = sc.nextInt();
            sc.nextLine();

            switch (choice) {

                case 1:
                    System.out.print("Enter username: ");
                    String regUser = sc.nextLine();
```

```
                System.out.print("Enter password: ");
                    String regPass = sc.nextLine();

                    service.register(regUser, regPass);
                    break;

            case 2:
                    System.out.print("Enter username: ");
                    String logUser = sc.nextLine();

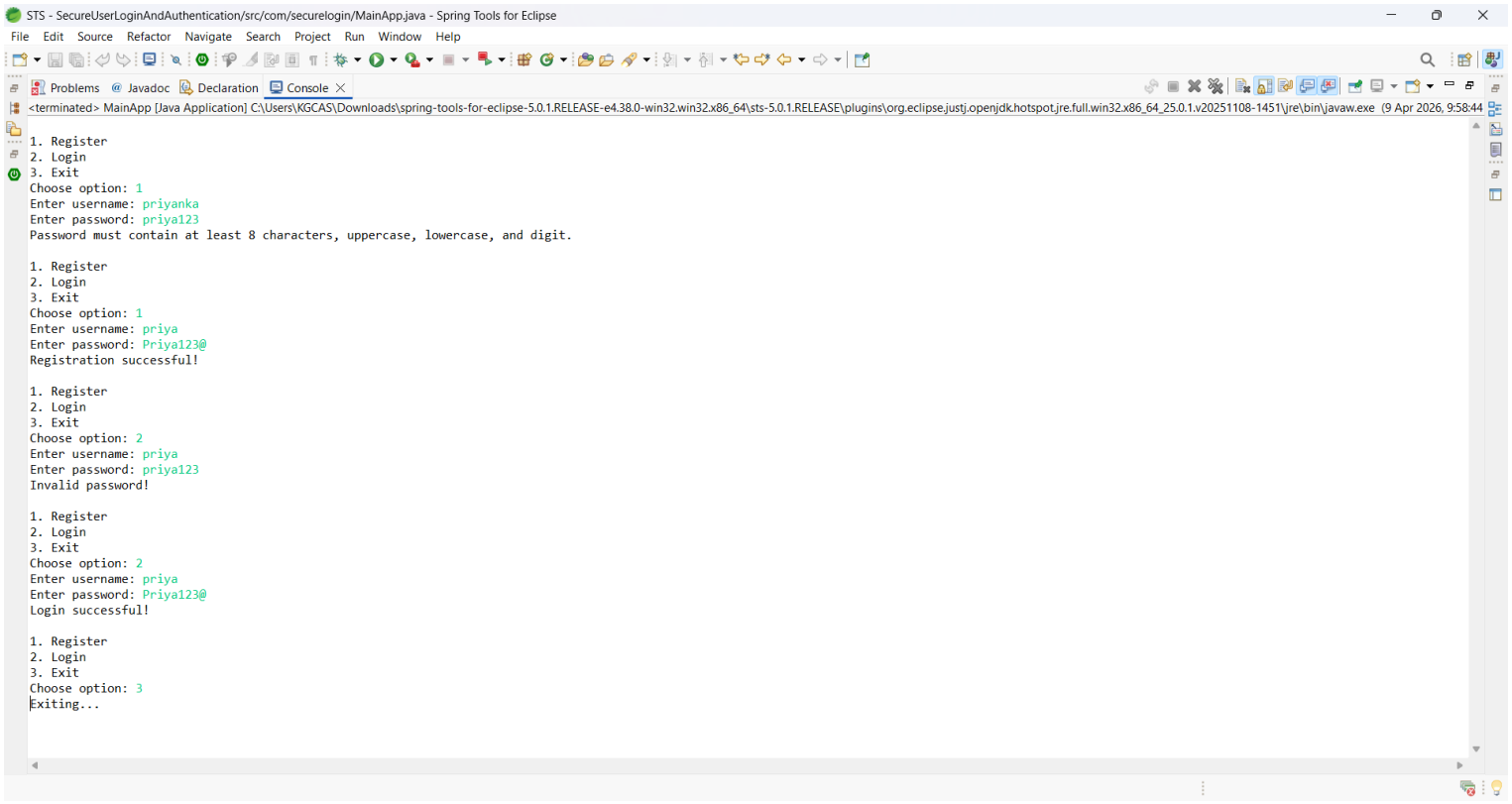
                    System.out.print("Enter password: ");
                    String logPass = sc.nextLine();

                    service.login(logUser, logPass);
                    break;

            case 3:
                    System.out.println("Exiting...");
                    sc.close();
                    System.exit(0);

                default:
                    System.out.println("Invalid choice!");
            }
        }
    }
```

6.INPUT AND OUTPUT SCREEN



7.DISSCUSION AND CONCLUSION

7.1 Discussion

The **Secure User Login and Authentication System** was developed to provide a secure user login and authentication mechanism with password validation. The system allows users to register with unique usernames and validated passwords, and log in securely.

During the development process, Core Java programming concepts such as loops, conditional statements, methods, regular expressions, and object-oriented principles were applied. The system successfully performs password validation and credential authentication.

The system was tested with different inputs including weak passwords, duplicate usernames, and wrong credentials to ensure it handles all scenarios correctly. The results show that the system effectively enforces password policies and prevents unauthorized access.

7.2 Conclusion

The **Secure User Login and Authentication System** successfully demonstrates the practical application of Core Java programming concepts in developing a real-world secure authentication system.

The system enforces password complexity rules, prevents unauthorized access, and provides clear authentication feedback. It simplifies secure credential management by validating and storing user information reliably.

This project helps students understand how Java programming can be used to develop secure authentication systems. The project can be further enhanced by adding features such as database connectivity for persistent storage, password hashing for security, graphical user interfaces, and session management.

8.REFRENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

Websites

• https://www.oracle.com/java
• https://www.geeksforgeeks.org
• https://www.w3schools.com
• https://www.javatpoint.com